ELEC 442 - A2

⇒ Notation:- $C_i = \cos\theta_i$, $S_i = \sin\theta_i$

$$\begin{bmatrix} V \\ \omega \end{bmatrix} = J\dot{q} \quad \text{(linear on top)}$$

→ all coordinates are wrt base frame $\mathcal{L}_0$.

1.) Key idea:-

$$J(:,i) = \begin{cases} \begin{bmatrix} k_{i-1} \\ 0 \end{bmatrix} & \text{joint } i \text{ prismatic} \\ \begin{bmatrix} k_{i-1} \times (O_6 - O_{i-1}) \\ k_{i-1} \end{bmatrix} & \text{joint } i \text{ revolute} \end{cases}$$

$$J(0) = I = [e_1 \ldots e_6].$$

a.) ~~*~~ Colouns 1-3 must equal $[e_i; 0]$: pure translation along $\underline{i}_0, \underline{j}_0, \underline{k}_0$ with no rotation

* (columns 4-6 must equal $[O_i; e_i]$:
rotation about $\underline{i}_0, \underline{j}_0, \underline{k}_0$ wrt K_{i-1} x
$(O_6 - O_{i-1}) = 0$. O_6 lies on every one of
these axes.

PPPRRR : Cartesian (gantry) arm
+ X-Y-Z Spherical wrist with wrist
Centre $= O_6$

→ Kinematics in exp. form :-

$$\underline{C}_6 = \underline{C}_0 e^{\theta_4 i^\times} e^{\theta_5 j^\times} e^{\theta_6 k^\times}$$

$$O_6 = O_0 + \underline{C}_0 \left(P_{home} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \right)$$

$${}^0T_6(\theta) = \begin{bmatrix} I & P_{home} \\ 0^T & 1 \end{bmatrix}$$

$\Rightarrow$ At home: $\underline{C}_6 = \underline{C}_0 \; (\underline{i}_6 = \underline{i}_0;\ \underline{k}_6 = \underline{k}_0)$

Polarities: d_1, d_2, d_3 Positive along $+\underline{i}_0, +\underline{j}_0, +\underline{k}_0$; $\theta_4, \theta_5, \theta_6$ right handed about their axes.

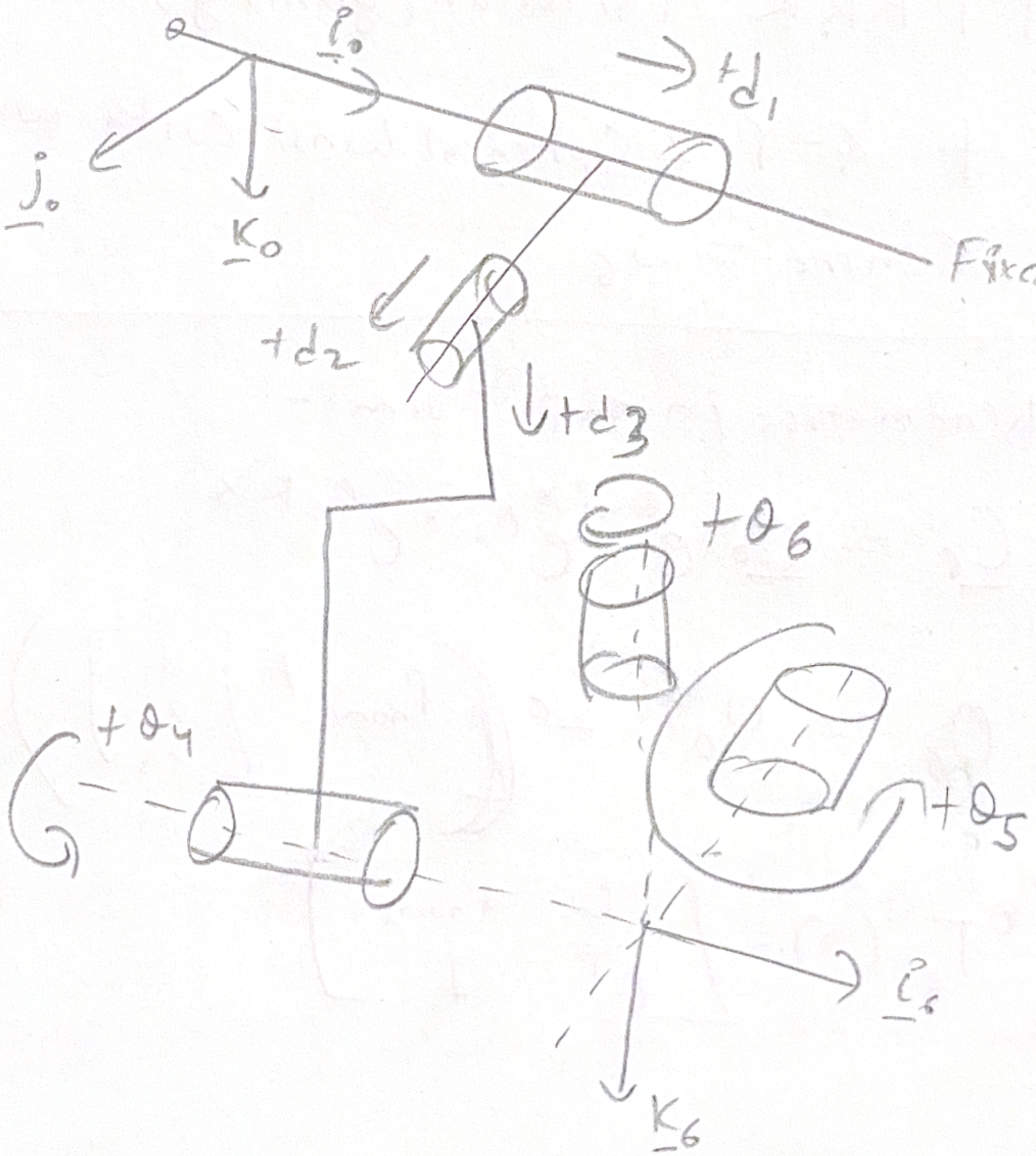

Fig: Home Position ($\underline{k}_0$ pointing down)

⇒ General Jacobian. The prismatic joints only translate, so $k_0, k_1, k_2 = i, j, k$ always, and the wrist axes (in $\mathcal{L}_0$ coordinates) are

$$k_3 = i, \quad k_4 = e^{\theta_4 i \times} j = \begin{bmatrix} 0 \\ c_4 \\ s_4 \end{bmatrix},$$

$$k_5 = e^{\theta_4 i \times} e^{\theta_5 j \times} k = - \begin{bmatrix} s_5 \\ -s_4 c_5 \\ c_4 c_5 \end{bmatrix}$$

$$\rightarrow J(q) = \begin{bmatrix} I_3 & 0_3 \\ 0_3 & J_{22} \end{bmatrix},$$

$$J_{22} = \begin{bmatrix} k_3 & k_4 & k_5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & s_5 \\ 0 & c_4 & -s_4 c_5 \\ 0 & s_4 & c_4 c_5 \end{bmatrix}$$

$$\Rightarrow \boxed{J(0) = I} \checkmark$$

b) Units of sub-matrices

with $\dot{q} = [\dot{d}_1\ \dot{d}_2\ \dot{d}_3\ (m/s);\ \dot{\theta}_4\ \dot{\theta}_5\ \dot{\theta}_6\ (rad/s)]$ and $[v\ (m/s);\ \omega\ (rad/s)]$:

J_{11}	$\dot{d} \to v$	dimensionless
J_{12}	$\dot{\theta} \to v$	m/rad
J_{21}	$\dot{d} \to \omega$	rad/m
J_{22}	$\dot{\theta} \to \omega$	dimensionless

c) Singular Configurations

$$\det J = \det J_{22} = 1 \cdot (C_4 C_4 S_5 + S_4 S_4 S_5)$$

$$= \cos(\theta_5)$$

$$\Rightarrow \boxed{\theta_5 = \pm \frac{\pi}{2} + n\pi \left[\pm 90°, \pm 270°, \ldots\right] \quad n \in \mathbb{Z}}$$

d.) Conditions giving $\bar{J} = I$

$J_{11} = I$ always, so we need $\bar{J}_{22} = I$:

$k_4 = j \Rightarrow c_4 = 1, s_4 = 0$; then

$k_5 = k \Rightarrow s_5 = 0, c_5 = 1.$

$$\theta_4 = 2\pi m, \theta_5 = 2\pi n \ (m, n \in \mathbb{Z})$$

0.)2.) D-H table From A1

=)

	θ_i (deg)	d_i (mm)	a_i (mm)	α_i (deg)
L1	θ_1	0	0	-90
L2	θ_2	0	431.8	180
L3	$\theta_3 + 90$	-149.1	20.3	90
L4	θ_4	433.1	0	90
L5	θ_5	0	0	-90
L6	θ_6	60	0	0

9.) Symbolic Jacobian & Singularities

→ All joints are revolute,

So $J(i,i) = \begin{bmatrix} k_{i-1} \times (O_6 - O_{i-1}) \\ k_{i-1} \end{bmatrix}$,

$i = 1, \ldots 6.$

⇒ Because $k_2 = -k_1$ ($\alpha_2 = 180°$) ⇒ Elbow enters as a difference :-

$$\theta_{23} = \theta_2 - \theta_3 \,, \quad r = a_2 c_2 + a_3 s_{23} + d_4 c_{23}$$

$$h = -a_2 s_2 + a_3 c_{23} - d_4 s_{23}$$

[Note:- $s_{23} = \sin(\theta_2 - \theta_3)$

$c_{23} = \cos(\theta_2 - \theta_3)$

$$\Rightarrow K_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, K_1 = -K_2 = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix},$$

$$K_3 = \begin{bmatrix} c_1 & c_{23} \\ s_1 & c_{23} \\ -s_{23} \end{bmatrix}, K_4 = \begin{bmatrix} -s_1 c_4 + c_1 s_{23} s_4 \\ c_1 c_4 + s_1 s_{23} s_4 \\ c_{23} s_4 \end{bmatrix},$$

$$K_5 = \begin{bmatrix} c_1 c_{23} c_5 - (s_1 s_4 + c_1 s_{23} c_4) s_5 \\ s_1 c_{23} c_5 + (c_1 s_4 - s_1 s_{23} c_4) s_5 \\ -s_{23} c_5 - c_{23} c_4 s_5 \end{bmatrix},$$

$$O_0 = O_1 = 0, O_2 = q_2 \begin{bmatrix} c_1 s_2 \\ s_1 c_2 \\ -s_2 \end{bmatrix}, P_w = O_4 = O_5 =$$

$$\begin{bmatrix} c_1 r + d_3 s_1 \\ s_1 r - d_3 c_1 \\ h \end{bmatrix}, O_6 = P_w + d_6 K_5.$$

⇒ decouple the wrist centre

$$\because \quad V_o = V_\omega + \omega \times (d_6 K_5),$$

$$\bar{J} = \begin{bmatrix} I & -d_6(K_5 \times) \\ 0 & I \end{bmatrix} \bar{J}_\omega$$

$$\bar{J}_\omega = \begin{bmatrix} \bar{J}_{11} & 0 \\ J_{21} & J_{22} \end{bmatrix}$$

$$\Rightarrow \det(\bar{J}) = -q_2 \sin\theta_5 \left(q_3 \cos\theta_3 + d_4 \sin\theta_3\right)$$

$$\cdot \left(q_2 \cos\theta_2 + q_3 \sin(\theta_2 - \theta_3) + d_4 \cos(\theta_2 - \theta_3)\right)$$

Note:- $\bar{J}_{21} = [K_0 \; K_1 \; K_2]$ $\bar{J}_{22} = [K_3 \; K_4 \; K_5]$

$$\rightarrow \bar{J}_{11} = \left[K_0 \times P_\omega \;\; K_1 \times P_\omega \;\; K_2 \times (P_\omega - O_2) \right]$$

i.) Wrist Singularity:- $\sin\theta_5 = 0$

$$\Rightarrow \boxed{\theta_5 = 0^\circ \text{ or } 180^\circ}$$

→ Then $K_3 \parallel K_5$; joints 4 & 6 produce the same rotation

ii.) Elbow Singularity: $a_3 C_3 + d_4 S_3 = 0$

$\Rightarrow \tan\theta_3 = -a_3/d_4$, $\boxed{\theta_3 = -2.68^\circ \text{ or } 177.32^\circ}$

→ The arm is Fully stretched / Folded, so the wrist centre cannot move along the arm

iii.) Shoulder Singularity: $a_2\cos\theta_2 + a_3\sin(\theta_2-\theta_3) + d_4\cos(\theta_2-\theta_3) = 0 \Rightarrow \boxed{r = 0}$

→ θ_1 can no longer move it sideways.

b) VH_homog.m File included

c) For Kin Poms 560.m File included.

d) Manipulability metrics

→ with singular values $\sigma_1 \geq 2 \cdots \geq \sigma_6$

of $\bar{J}$: manipulability $\omega = \sqrt{\det(J\bar{J}^T)}$

$= \prod \sigma_i$, minimum

Singular value σ_{min}, and Condition number

$K = \sigma_{max}/\sigma_{min}$

Joint set	ω	σ_{min}	K
q_A	0	0	∞
q_B	0	0	∞
q_C (Jin mm)	1.973×10^{7}	0.387	2501
q_C (Jin m)	1.973×10^{-2}	0.137	13.9

* q_A & q_B both have $\theta_5 = 0$ and are wrist singular: $\omega = \sigma_{min} = 0$, $K = \infty$.

In q_B row 4 of J is entirely zero, no joint-rate combination produces ω_n.

Neither pose is at elbow or shoulder singularity.

* q_C is well away from all three conditions ($s_5 = -0.5$, elbow $= 320.6$ mm, $r = 285.0$ mm),

so $\omega > 0$, $\sigma_{min} > 0$.

* These values of unit dependent.

Only comparisons in the same units are meaningful; a fair isotropy measure needs J normalised.

e) Figures have been added with the File.